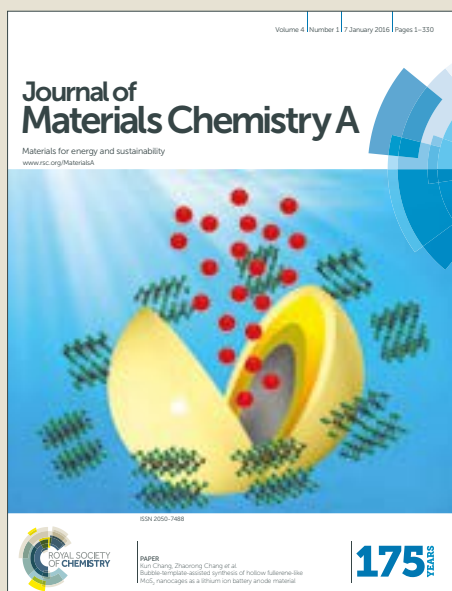


Journal of Materials Chemistry A

Accepted Manuscript



This article can be cited before page numbers have been issued, to do this please use: W. Ying, Q. Hou, Y. Guo, D. Chen, Z. Li, J. Zhang, Y. Yan and X. Peng, *J. Mater. Chem. A*, 2019, DOI: 10.1039/C9TA01636H.



This is an Accepted Manuscript, which has been through the Royal Society of Chemistry peer review process and has been accepted for publication.

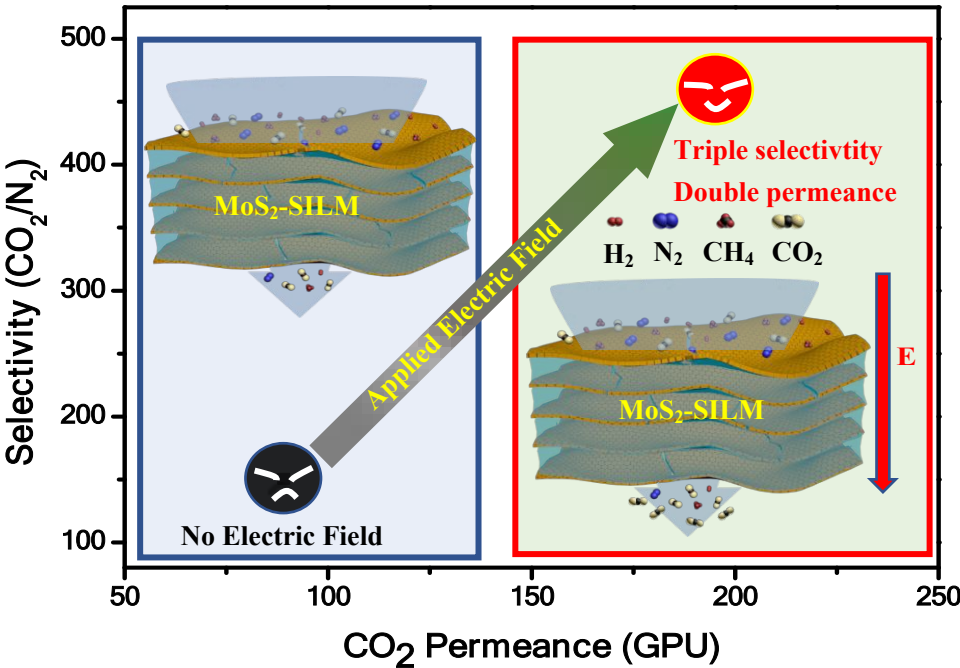
Accepted Manuscripts are published online shortly after acceptance, before technical editing, formatting and proof reading. Using this free service, authors can make their results available to the community, in citable form, before we publish the edited article. We will replace this Accepted Manuscript with the edited and formatted Advance Article as soon as it is available.

You can find more information about Accepted Manuscripts in the [author guidelines](#).

Please note that technical editing may introduce minor changes to the text and/or graphics, which may alter content. The journal's standard [Terms & Conditions](#) and the ethical guidelines, outlined in our [author and reviewer resource centre](#), still apply. In no event shall the Royal Society of Chemistry be held responsible for any errors or omissions in this Accepted Manuscript or any consequences arising from the use of any information it contains.

Table of content

View Article Online
DOI: 10.1039/C9TA01636H



Electric field dramatically improves the permeance and selectivity of CO₂ over H₂, CH₄, N₂ through MoS₂ supported ionic liquid membrane (MoS₂-SILM).

ARTICLE

Electrical field facilitates selectively transporting CO₂ through laminated MoS₂ supported ionic liquid membrane

Wen Ying,^{a†} Quangang Hou,^{b†} Danke Chen,^a Yi Guo,^a Zhuoyi Li,^a Jun Zhang,^b Youguo Yan,^{b*} and Xinsheng Peng^{a*}

Received 00th January 20xx,
Accepted 00th January 20xx

DOI: 10.1039/x0xx00000x

Ionic liquid (IL), a neutral assembly of charged ions, is a potential material for CO₂ capture and storage. The practicability of IL can be improved by confining it into porous substrate and nanopores, which will further enhance the performance of IL by changing its structure. On the other side, external electric field (EEF) should also influence IL structure through its charged component. In this work, we apply EEF on MoS₂ supported ionic liquid membrane (MoS₂-SILM) to facilitate selectively transporting CO₂ by dual influence of MoS₂ nanochannel and EEF. The redistribution of nanoconfined IL under EEF results to the increase of free volume of IL and decrease of anion-cation interaction, which lead to the enhancement of CO₂ solubility and diffusion, and further, the permeance. Compared with the MoS₂-SILM without EEF, CO₂ permeance is enhanced from 89 GPU to 200 GPU, and the selectivity of CO₂/H₂, CO₂/CH₄ and CO₂/N₂ is also enlarged 2~3 times. The interesting phenomenon is the asymmetric CO₂ separation performance influenced by the CO₂ absorbed on MoS₂ nanosheets under positive and negative EEF, and may open a new strategy to design high performance gas separation membrane.

1 Introduction

The massive emission of CO₂, a kind of greenhouse gas, is thought to be a significant cause of global warming and other climate anomalies, so efficient carbon capture and storage is an emergency demand for environmental protection and sustainable development.¹ The existing methods, such as amine-based absorption, porous solids-based adsorption, and cryogenic distillation, both rely on high energy-consumption or high cost.²⁻⁴ Therefore, more efficient and economical strategy and material are required.

Ionic liquid (IL), salt of organic cation and inorganic or organic anion, is considered as a promising material for CO₂ capture and storage because its low volatility, high thermal stability and tuneable physicochemical property.⁵ Experiment has confirmed that conventional IL exhibits high CO₂ solubility, especially imidazolium-based IL,⁶ and can further be improved by increasing the alkyl chain length or fluoroalkyl chain.^{7, 8} However, the physical absorption of CO₂ in conventional IL isn't practical as it usually needs high pressure and a certain time to equilibrate.⁹ So, functionalized IL that can both physical and chemical adsorb CO₂ comes out.¹⁰⁻¹² Nevertheless, these task-specific IL always brings high viscosity, which is stumbling block

in industrial practice,¹³ and it also needs large amount of IL if directly using IL to absorb CO₂ as liquid.¹⁴

To overcome the above drawback, confining IL into porous substrate to produce supported ionic liquid membrane (SILM) is a practical method.¹⁵ But the unavoidable leakage under high pressure still limits the usage, which can lead to two solutions, that are thicker membrane and smaller pore size. However, thicker membrane is always accompanied by lower permeance, so smaller pore size is the better choice. When the pore size comes to nanometer scale, usually several nanometers, IL will be nanoconfined and exhibit different property,¹⁶ such as anomalous thermal property¹⁷ and redistribution of ions,^{18, 19} which may have benefit in CO₂ separation. Much work has confirmed that nanoconfined IL has an increase in CO₂ adsorption or CO₂ diffusion.^{20, 21} Except that, using nanoconfined IL can reduce the amount of required IL and take advantage of its high viscosity to lift stability. Many materials with nanopores or nanochannels like metal organic frameworks,^{22, 23} zeolite,^{24, 25} silica²⁶ and two-dimension material²⁷⁻²⁹ can be used as porous substrate to confine IL and excellent CO₂ separation performance has been achieved. However, there is a limitation in performance improvement as the limited change of IL in nanoconfinement, and the redistribution of IL cannot be further tuned as soon as IL being nanoconfined. So more flexible and stronger force is needed to drive the redistribution of IL.

External stimuli, such as temperature, pressure and electric field is always used to break the intrinsic ceiling. As IL is liquid electrolyte made of ionic molecules, external electric field (EEF) should have influence on its anion and cation, like forming electric double layer^{30, 31} or generating electrorheological effect³². It has been confirmed that anion and cation will

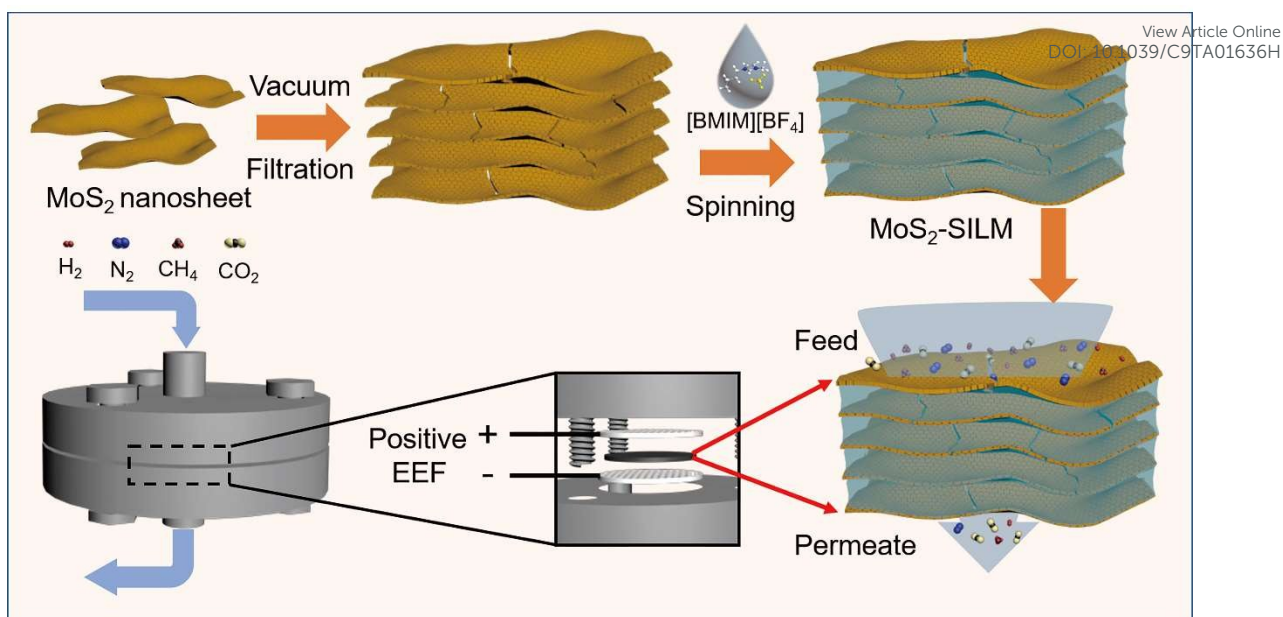
^a State Key Laboratory of Silicon Materials, Department of Materials Science and Engineering, Zhejiang University, Hangzhou 310027, China.

^b College of Science, China University of Petroleum (East China), Qingdao, China, 266580.

[†] W. Y. and Q. H. contributed equally to this work.

Correspondence and requests for materials should be addressed to X. P. (email: pengxinsheng@zju.edu.cn), or Y. Y. (email: yyg@upc.edu.cn).

Electronic Supplementary Information (ESI) available: [details of any supplementary information available should be included here]. See DOI: 10.1039/x0xx00000x



Scheme 1 The preparation process and gas separation of MoS₂-SILM by home-made separation device.

directionally move under EEF and this redistribution can also change the property of IL.³³ Inspired by it, we guess that applying EEF on nanoconfined IL should have dual influence on IL structure and may further improve CO₂ separation performance.

In this work, we apply EEF on molybdenum disulfide supported ionic liquid membrane (MoS₂-SILM) to tune the redistribution of IL and achieve the controllable performance. Anion and cation of [BMIM][BF₄] under EEF will move to cathode and anode, respectively. The degree of movement can be easily controlled by the docile EEF. The redistribution of [BMIM][BF₄] changes the anion-cation interaction and free volume of IL and thus improves the permeance of CO₂ from 89 GPU to 200 GPU, with 2~3 times of the selectivity for CO₂/H₂, CO₂/CH₄ and CO₂/N₂ pairs, respectively.

2 Results and discussion

2.1 Synthesis and characterization of MoS₂-SILM.

The MoS₂ nanosheet was exfoliated by n-butyllithium, the same process as our previous work about MoS₂-SILM²⁸ and details can be seen in Method section. AFM images (Fig. S1) show the size of single MoS₂ nanosheets is about several micrometer with the average height of 1.2 nm. MoS₂ membrane was prepared by vacuum filtration of MoS₂ aqueous dispersion (Scheme 1) on anodic alumina oxide (AAO) and the morphology is shown in Fig. 1a and 1c. There are many short and thick wrinkles on the surface and regular layered structure on the cross section, which is caused by the stacking of MoS₂ nanosheets. As MoS₂ has good wettability of [BMIM][BF₄],²⁸ [BMIM][BF₄] can be easily trapped into MoS₂ nanochannel by capillary force while dropped on the surface of MoS₂ membrane. After spinning

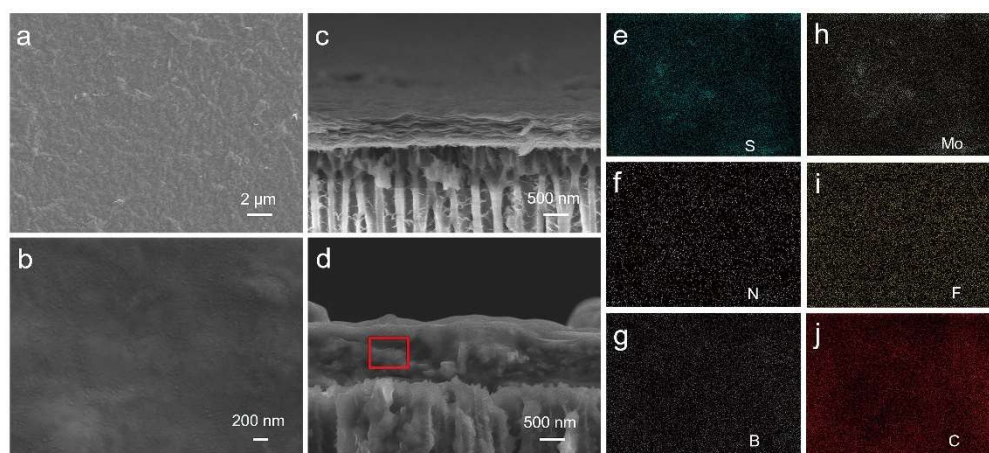


Fig. 1 The SEM images of MoS₂ membrane and MoS₂-SILM and EDS spectrum of MoS₂-SILM. (a) and (b) are surface of MoS₂ and MoS₂-SILM; (c) and (d) are cross section of MoS₂ and MoS₂-SILM; (e)-(h) are the EDS mapping images of Mo, S, B, N, F, C recorded from the red pane in d, respectively.

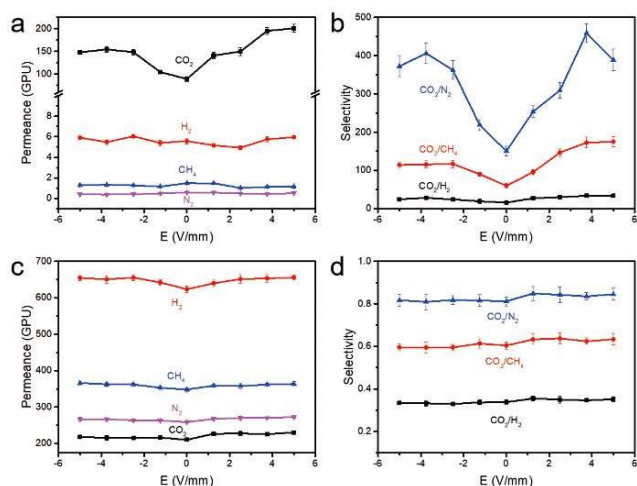


Fig.2 Gas separation performance of MoS₂ membrane and MoS₂-SILM. (a) and (b) are the permeance and selectivity of MoS₂-SILM membrane under EEF; (c) and (d) are the permeance and selectivity of MoS₂ membrane under EEF; Error is within $\pm 15\%$.

away the residue [BMIB][BF₄] on MoS₂ surface, MoS₂-SILM was prepared. Wrinkles on the surface and layered structure on the cross section of MoS₂ membrane were not observed in MoS₂-SILM. The thickness is expanded from 540 nm of MoS₂ membrane to 1120 nm of MoS₂-SILM, which demonstrate the combination of MoS₂ membrane and [BMIB][BF₄] (Fig.1b and 1d). The mapping images of MoS₂-SILM's cross section are shown in Fig. 1e~j. The uniform distribution of B, N, and F elements confirms the uniform distribution of [BMIB][BF₄] in MoS₂-SILM. The FTIR spectra of MoS₂ membrane, [BMIB][BF₄] and MoS₂-SILM are shown in Fig. S2 and Table S1. The obvious peak shifts of MoS₂-SILM compared with bulk [BMIB][BF₄] means the interaction between MoS₂ nanosheets and [BMIB][BF₄], which confirms the nanoconfinement of [BMIB][BF₄].²⁸

2.2 Gas separation.

IL is made of charged ionic, so both anion and cation of IL will directionally move under electric field. In our work, we applied EEF on MoS₂-SILM by a home-made device as shown in Scheme 1, and the capacitor plates were not directly put on membranes to avoid Joule effect.³⁴ Gas permeance under EEF was tested by bubble flow meter with 0.01 ml accuracy, under 0.1 MPa and 293 K, and the result is shown in Fig. 2. CO₂ permeance has obviously increase both under positive and negative EEF with the largest permeance of 200 GPU. But the magnitude of increase under positive EEF is larger than negative EEF resulting

in asymmetry permeance. On the contrary, EEF doesn't have obvious effect on the permeance of H₂, CH₄ and N₂ through MoS₂-SILM. H₂ permeance decreases and sometimes slightly increases with the strengthening of EEF, while for CH₄ and N₂, permeance has little decrease both under positive and negative EEF. As a result, gas selectivity of CO₂/H₂, CO₂/CH₄ and CO₂/N₂ is in accord with CO₂ permeance, and shows a certain degree of asymmetry, which increases under EEF with different increase magnitude under positive and negative EEF. As a comparison, we also measured the performance of MoS₂ membrane under EEF. Fig. 2c and 2d shows that all the gases permeance through MoS₂ membrane has a little increase under positive and negative EEF, and the selectivity is still close to Knudsen selectivity³⁵ without obvious change. Above all, EEF greatly improves the performance of MoS₂-SILM for separating CO₂ over H₂, CH₄ and N₂, and surpass Robeson Upper bound (Fig. S3) and most of other membranes (Fig. 3 and Table S2, detail references can be seen in Supporting information). Mixed gas separation was also tested by gas chromatography and the tendency is in accord with the ideal selectivity. However, the values and magnitude of change are smaller than the corresponding ideal selectivity, as shown in Fig. S4.

Basing on the above experimental phenomenon, we processed the molecular dynamics (MD) simulation to investigate the mechanism of EEF enhanced CO₂ separation performance through MoS₂-SILM. Fig. S5 shows, the computational CO₂ permeance increases with the increasing electric field intensity both under positive and negative EEF, while the increase under positive EEF is more significant. This result is consistent with the experimental result. Further, the IL structure in nanochannel is investigated. First, we plotted the number density profile of [BMIM]⁺ and [BF₄]⁻ in the nanochannel without EEF along the thickness direction in Fig. 4a. The nanoconfined IL demonstrates a layered structure and the density of primary adsorption layers is much larger than that of central layers. But the layered structure of [BMIM]⁺ and [BF₄]⁻ has a little difference. The two primary peaks of cations are higher than that of the anions, which can be attributed to the fact that the negative charged sulfur atoms (on the surface of molybdenum disulfide) have a higher attraction to the positively charged cations. Moreover, the primary adsorption layer nears the feed side, is slightly higher than that of the primary adsorption layer nears the permeate side, for both cations and

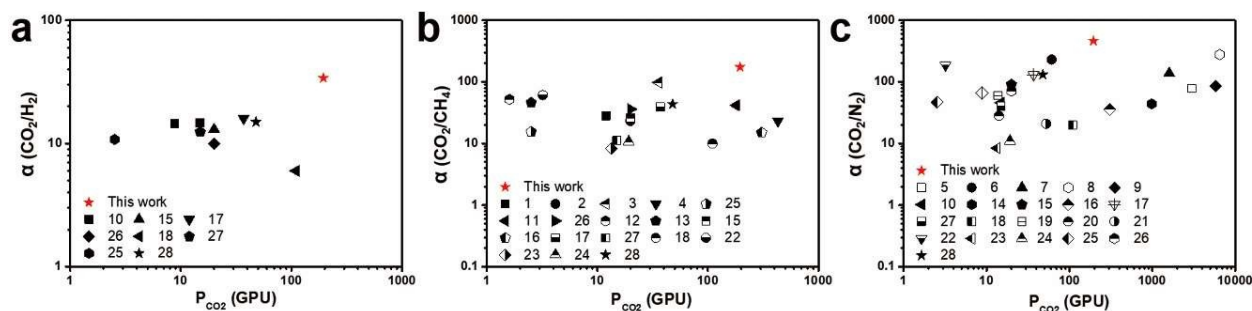


Fig. 3 The separation performance (selectivity vs. permeance) of MoS₂-SILM under EEF compared with other membranes. (a) CO₂/H₂, (b) CO₂/CH₄ and (c) CO₂/N₂. The detail data is shown in Table S2 and from the Supplementary references.

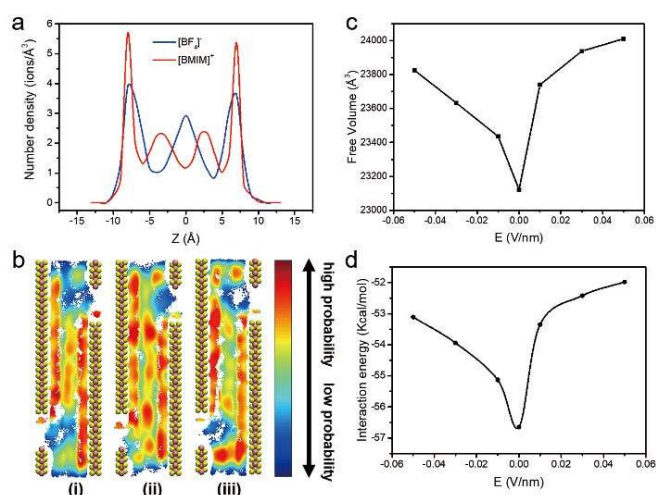


Fig. 4 (a) is the number density profile of [BMIM]⁺ and [BF₄]⁻ in the nanochannel without EEF along the thickness direction, and Z=0 represents the midline of MoS₂ nanochannel; (b) is [BF₄]⁻ density map in nanochannel under EEF of (i) E=-0.03 V/nm, (ii) E=0 V/nm, (iii) E=0.03 V/nm respectively. Blue indicates the region that low probability to find a [BF₄]⁻, while red region indicates the high probability of finding an anion; (c) is molecular simulated free volume of IL under EEF; (d) is computed [BMIM]⁺ and [BF₄]⁻ interaction energy as a function of EEF.

anions, which is due to the strong interaction between IL and those CO₂ adsorbed on the feed side MoS₂ nanosheets. In that case, the adsorbed CO₂ molecules make the IL more easily move to the feed side of the nanochannel.

EEF will drive these cations and anions of IL to move aside and change their adsorption structure inside nanochannel. As most of the [BMIM]⁺ is tightly adsorbed on MoS₂ surface and its large molecule weight makes it harder to move than [BF₄]⁻, the adsorption structure of [BMIM]⁺ changes little, its density map confirms this point (see Fig. S6). The density maps of [BF₄]⁻ under different EEF are also calculated, as shown in Fig. 4b. As to the [BF₄]⁻ layer adhere to MoS₂ surface, the strong adsorption guarantees its stable adsorption under EEF. In contrast, the central [BF₄]⁻ layer has high mobility capability subjected to EEF. Under positive and negative EEF, the central [BF₄]⁻ will move to feed and permeate side, respectively. Furthermore, under positive EEF, the shifted [BF₄]⁻ will be attracted by those adsorbed CO₂ on the feed side, and the deviation degree of central [BF₄]⁻ layer is high, while is just the reverse under negative EEF. Consequently, the structure is looser under positive EEF than that under negative EEF, confirmed by simulated free volume of IL under EEF as shown in Fig. 4c, and thus giving rise to higher CO₂ solubility and further to higher CO₂ permeance.³⁶

What's more, the movement of anion and cation under EEF will influence cation-anion interaction energy (Fig. 4d), which determined the IL viscosity and closely affects CO₂ permeance. Since the cations and anions subjected to converse electric field force, [BMIM]⁺ and [BF₄]⁻ prone to be departed under EEF, resulting in decrease of cation-anion interaction energy, and decreasing extent increases with strengthening electric field intensity. As to the profile asymmetry of Fig. 4d, it could be ascribed to the additional attraction between CO₂ and [BF₄]⁻ under positive EEF (mentioned above), which results in an increase of cation-anions separation degree and yields

weakened interaction. Previous experiments and simulation results both indicated that the weak cation-anion interaction determines a low viscosity³⁷ and allows a high CO₂ diffusion coefficient,³⁸ which yielding to a high CO₂ permeance. Therefore, the weakened cation-anion interaction could be also used to rationalize the improved CO₂ permeance under EEF.

Our simulation results indicate that the enhanced CO₂ permeability in MoS₂-SILM under EEF could be ascribed to the IL structure change inside nanochannel, especially the small-size [BF₄]⁻. Under EEF, the loosed IL structure and decreased cation-anion interaction is responsible for the enhanced CO₂ permeance. The asymmetric behavior of CO₂ under positive and negative EEF is derived of the attraction between IL and adsorbed CO₂ on MoS₂ nanosheet adjacent to feed side. To be specific, CO₂ molecules that adsorbed on the feed side of the nanochannel makes the anions more easily move to the feed side of the nanochannel under positive EEF while restrict the move under negative EEF, which causes the asymmetric increase of free volume and decrease of cation-anion pair attraction.

It is an interesting phenomenon that CO₂ can also influence the distribution of IL in MoS₂ nanochannel. As we know, functionalizing the transportation channel and applying external stimuli are two common methods to improve the performance of gas separation membrane. However, our results indicate that the structure of membrane can future be tuned by target gas to form a specific but fast transportation channel.

The time durability and electric field durability of the resulted MoS₂-IL membrane is shown in Fig. S7. It indicates that the CO₂ permeance and selectivity has little change after testing for a week under 5V/mm. However, the performance degraded continuously when EEF reaches to 12.5 V/mm due to the leakage of IL (Fig. S8), as confirmed that after testing under 12.5 V/mm, the AAO support was filled with IL, exuded from MoS₂-SILM.

3 Method

3.1 Synthesis of MoS₂ membrane and MoS₂-SILM.

MoS₂ nanosheets were prepared by exfoliating MoS₂ powder, purchased from Aladdin, in a glove box, as reported elsewhere.²⁸ 200 mg MoS₂ powder was dispersed in 2 ml hexane solution of n-butyl lithium (1.6 M) with stirring for 48 h under room temperature. Then the mixture was filtered on a PTFE filter membrane and washed by enough hexane. The obtained powder was dispersed in 100 ml deionized water under ultrasonication for 1 h and followed by centrifuging. The supernatant was just the MoS₂ nanosheets dispersion. MoS₂ membrane was prepared by vacuum filtration 3 ml MoS₂ nanosheets dispersion (0.3 mg/ml) on an AAO substrate. 0.2 ml [BMIM][BF₄] was dropped on the resultant MoS₂ membrane and kept for 12 h, followed by spinning with 1000 r/m for 6 s and 7000 r/m for 60 s.

3.2 Characterization.

SEM images were obtained by Hitachi S-4800 field-emission SEM equipped with an EDS spectrometer. AFM images was

obtained by Bruker Dimension Edge and FTIR spectrum was obtained by Bruker Tensor 27.

Gas Separation. Bubble flow meter with total volume of 1 ml with 0.01 ml accuracy was used to measure gas permeance. The MoS₂-SILM was assembled in an in-line stainless steel filter holder with effective membrane surface area was 2.2 cm², purchased from Millipore. The gas permeance measurements were performed ten times for each membrane and three membranes were evaluated. Gas permeance (*Q*) was calculated according to the equation (1):²⁸⁻²⁹

$$Q = \frac{1}{P_u - P_d} * \frac{273.15}{273.15 + T} * \frac{P_{atm}}{101.325} * \frac{1}{A} * \frac{dV}{dt} \quad (1)$$

where *P_u* is the upstream pressure, *P_d* is the downstream pressure (atmosphere in our testing environment), *P_{atm}* is the atmospheric pressure, *A* is the membrane effective area, *T* is the temperature, *dV/dt* is the volumetric displacement rate in the bubble flow meter. The 50% porosity of the AAO was considered. The ideal selective factor of two gases is the ratio of two gas permeance:²⁸⁻²⁹

$$\alpha = \frac{Q_1}{Q_2} \quad (2)$$

where *Q₁* and *Q₂* are the two permeance, respectively.

The gas permeability was calculated by multiplying permeance (*Q*) and thickness.

The original mixed gas CO₂/H₂, CO₂/CH₄ and CO₂/N₂ were 49.6% : 50.4%, 50.1%:49.9% and 49.7%:50.3% in volume ratio, respectively, and mixed gas separation factor was measured by FULI 9790 gas chromatograph under 293 K and 0.1 Mpa.

3.3 Molecular dynamic simulation.

3.3.1 Model system. A bichamber system is set up to study the effect of electric field on the CO₂ permeability of MoS₂-SILM, as shown in Fig. S5. Two MoS₂ sheets were used to build a nanochannel with interlayer spacing of 2.5 nm (measured in the experiment). Each MoS₂ sheets contains a notch of 1.9 nm, and the nanochannel offset is 3.8 nm. The nanochannel was filled with 180 [BMIM][BF₄] ionic liquid pairs. Two rigid helium sheets were placed at the rightmost and leftmost sides of the system. 650 CO₂ molecules on the left side was used as a feed chamber with an initial pressure of 6 MPa. A vacuum layer with thickness of 160 Å on the other side was used as the permeate chamber. The simulation box has dimensions of 50 Å × 100 Å × 300 Å. Periodic boundary conditions were used in all three directions. The direction from left to right is defined as the positive direction of electric field. In simulation, all these MoS₂ and graphene sheets were fixed. A series of electric fields with intensity of 0.01~0.03 V/nm for both positive and negative directions were applied to study electric field influence on CO₂ permeance. Higher electric field intensity in simulation than experiments were applied in order to obtain more precise statistics for the time scales involved in our molecular dynamic simulation.

3.3.2 Molecular dynamics simulations. All the molecular dynamics (MD) simulations were carried out with the LAMMPS package. During the simulation, non-polarizable all-atom OPLS-AA force field was used for IL. The Lennard-Jones parameters

and atom charge for the [BMIM][BF₄] and CO₂ came from our previous work.²⁹ The Lennard-Jones parameters and atom charge for the MoS₂ were obtained from Ref. [39]³⁹. The cut-off distance for all the short-range vdW interactions was set as 12 Å. The long-range electrostatic interactions were computed by using the particle-particle particle-mesh (PPPM) algorithm. All the data reported in this work were averaged over a set of 5 simulations from various initial velocity distributions. For each system, the simulation was conducted in a canonical ensemble (NVT) at 298 K controlled by the Nose-Hoover thermostat method. The time step was set as 1 fs, and the data were collected every 1 ps. The total simulation time of each model was 10 ns.

4 Conclusions

EEF can redistribute anion and cation of IL, which is nanoconfined in MoS₂ channel, and thus greatly improve the performance of MoS₂-SILM, with CO₂ permeance from 89 to 200 GPU and selectivity of CO₂/H₂, CO₂/CH₄ and CO₂/N₂ from 16, 60, 150 to 34, 174 and 462, respectively. The interaction between EEF and IL, combined with the interaction between adsorbed CO₂ and anion leads to the interesting asymmetry of CO₂ permeance, which may open a new method to functionalize gas separation membrane.

Conflicts of interest

There are no conflicts to declare.

Acknowledgements

This work is supported by the National Natural Science Foundations of China (NSFC 21671171, 21875212, 51632008), the National Basic Research Program of China 973 Program (2015CB655302), National Key Research and Development Program (2016YFA0200204), and the Zhejiang Natural Science Foundation (LD18E02001).

Notes and references

1. M. Z. Jacobson, *Energy Environ. Sci.*, 2009, **2**, 148-173.
2. A. Goeppert, M. Czaun, G. K. S. Prakash and G. A. Olah, *Energy Environ. Sci.*, 2012, **5**, 7833-7853.
3. M. George and R. G. Weiss, *J. Am. Chem. Soc.*, 2001, **123**, 10393-10394.
4. G. T. Rochelle, *Science*, 2009, **325**, 1652-1654.
5. S. Zeng, X. Zhang, L. Bai, X. Zhang, H. Wang, J. Wang, D. Bao, M. Li, X. Liu and S. Zhang, *Chem. Rev.*, 2017, **117**, 9625-9673.
6. J. E. Brennecke and B. E. Gurkan, *J. Phys. Chem. Lett.*, 2010, **1**, 3459-3464.
7. J. L. Anthony, J. L. Anderson, E. J. Maginn and J. F. Brennecke, *J. Phys. Chem. B*, 2005, **109**, 6366-6374.
8. M. J. Muldoon, S. N. V. K. Aki, J. L. Anderson, J. K. Dixon and J. F. Brennecke, *J. Phys. Chem. B*, 2007, **111**, 9001-9009.
9. J.-W. Ma, Z. Zhou, F. Zhang, C.-g. Fang, Y.-T. Wu, Z.-B. Zhang and A.-M. Li, *Environ. Sci. Technol.*, 2011, **45**, 10627-10633.

10. I. Niedermaier, M. Bahlmann, C. Papp, C. Kolbeck, W. Wei, S. K. Calderon, M. Grabau, P. S. Schulz, P. Wasserscheid, H.-P. Steinrueck and F. Maier, *J. Am. Chem. Soc.*, 2014, **136**, 436-441.
11. X. Luo, Y. Guo, F. Ding, H. Zhao, G. Cui, H. Li and C. Wang, *Angew. Chem., Int. Ed.*, 2014, **53**, 7053-7057.
12. G. Gurau, H. Rodriguez, S. P. Kelley, P. Janiczek, R. S. Kalb and R. D. Rogers, *Angew. Chem., Int. Ed.*, 2011, **50**, 12024-12026.
13. M. D. Soutullo, C. I. Odom, B. F. Wicker, C. N. Henderson, A. C. Stenson and J. H. Davis, Jr., *Chem. Mater.*, 2007, **19**, 3581-3583.
14. Y. Gu and G. Li, *Adv. Synth. Catal.*, 2009, **351**, 817-847.
15. L. J. Lozano, C. Godínez, A. P. de los Ríos, F. J. Hernández-Fernández, S. Sánchez-Segado and F. J. Alguacil, *J. Membr. Sci.*, 2011, **376**, 1-14.
16. S. Zhang, J. Zhang, Y. Zhang and Y. Deng, *Chem. Rev.*, 2017, **117**, 6755-6833.
17. S. Chen, K. Kobayashi, Y. Miyata, N. Imazu, T. Saito, R. Kitaura and H. Shinohara, *J. Am. Chem. Soc.*, 2009, **131**, 14850-14856.
18. V. V. Chaban and O. V. Prezhdo, *Acs Nano*, 2014, **8**, 8190-8197.
19. Y. Shim and H. J. Kim, *Acs Nano*, 2009, **3**, 1693-1702.
20. J. Zhang, Q. Zhang, X. Li, S. Liu, Y. Ma, F. Shi and Y. Deng, *Phys. Chem. Chem. Phys.*, 2010, **12**, 1971-1981.
21. W. Shi and D. R. Luebke, *Langmuir*, 2013, **29**, 5563-5572.
22. K. M. Gupta, Y. Chen and J. Jiang, *J. Phys. Chem. C*, 2013, **117**, 5792-5799.
23. Y. Ban, Z. Li, Y. Li, Y. Peng, H. Jin, W. Jiao, A. Guo, P. Wang, Q. Yang, C. Zhong and W. Yang, *Angew. Chem., Int. Ed. Engl.*, 2015, **54**, 15483-15487.
24. Y. Yu, J. Mai, L. Wang, X. Li, Z. Jiang and F. Wang, *Sci. Rep.*, 2014, **4**, 5997.
25. B. Yang, F. Zhou, S. Liu, P. Wang, A. S. Alshammari and Y. Deng, *RSC Adv.*, 2015, **5**, 48908-48915.
26. S. Tanaka, K. Kida, H. Fujimoto, T. Makino and Y. Miyake, *Langmuir*, 2011, **27**, 7991-7995.
27. W. Ying, J. Cai, K. Zhou, D. Chen, Y. Ying, Y. Guo, X. Kong, Z. Xu and X. Peng, *ACS Nano*, 2018, **12**, 5385-5393.
28. D. Chen, W. Ying, Y. Guo, Y. Ying and X. Peng, *ACS Appl. Mater. Interfaces*, 2017, **9**, 44251-44257.
29. D. Chen, W. Wang, W. Ying, Y. Guo, D. Meng, Y. Yan, R. Yan and X. Peng, *J. Mater. Chem. A*, 2018, **6**, 16566-16573.
30. M. T. Camci, P. Aydogan, B. Ulgut, C. Kocabas and S. Suzer, *Phys. Chem. Chem. Phys.*, 2016, **18**, 28434-28440.
31. M. V. Fedorov and A. A. Kornyshev, *Chem. Rev.*, 2014, **114**, 2978-3036.
32. Y. Liu, B. Wang, Y. Dong, X. Zhao and J. Yin, *J. Phys. Chem. B*, 2018, **122**, 12184-12193.
33. W. Sitaputra, D. Stacchiola, J. F. Wishart, F. Wang and J. T. Sadowski, *Adv. Mater.*, 2017, **29**.
34. J. E. Garay, U. Anselmi-Tamburini and Z. A. Munir, *Acta Mater.*, 2003, **51**, 4487-4495.
35. M. Knudsen, *Ann. Phy-Berlin*, 1909, **29**, 75-130.
36. M. S. Shannon, J. M. Tedstone, S. P. O. Danielsen, M. S. Hindman, A. C. Irvin and J. E. Bara, *Ind. Eng. Chem. Res.*, 2012, **51**, 5565-5576.
37. T. Kavitha, T. Vasanth, P. Venkatesu, R. S. R. Devi and T. Hofman, *J. Mol. Liq.*, 2014, **198**, 11-20.
38. X. Zhang, X. Zhang, H. Dong, Z. Zhao, S. Zhang and Y. Huang, *Energy Environ. Sci.*, 2012, **5**, 6668.
39. T. Liang, S. R. Phillpot and S. B. Sinnott, *Phys. Rev. B*, 2009, **79**.

View Article Online
DOI: 10.1039/C9TA01636H